

JSS 2 MATHEMATICS – FIRST TERM

TOPIC 2: BASIC OPERATION OF INTEGERS / WHOLE NUMBERS II – PRIME FACTORS & OPERATIONS

1. Prime Numbers and Prime Factors

A prime number has exactly two distinct positive divisors: 1 and itself (e.g. 2, 3, 5, 7, 11, 13). A prime factor is a factor of a number that is prime. Every whole number greater than 1 can be written as a product of prime factors (fundamental theorem of arithmetic).

2. Finding Prime Factors

Use successive division by the smallest primes.

Example: $60 = 2 \times 2 \times 3 \times 5 = 2^2 \times 3 \times 5$

Example: $84 = 2^2 \times 3 \times 7$

3. Applications

Prime factors are used to find LCM and HCF, to simplify fractions and to solve some quantitative reasoning problems.

4. Operations on Whole Numbers

Review addition, subtraction, multiplication and division of whole numbers. Pay attention to place value and order of operations (BODMAS/BIDMAS).

5. Summary

Prime factors are the building blocks of whole numbers. Expressing numbers as products of primes helps with LCM, HCF and other calculations. Master the four basic operations with accuracy.

End of Topic 2